TED [15]-3182
[REVISION 2015]

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/ TECHNOLOGY

(MODEL QUESTION PAPER)

SURVEYING FOR ARCHITECTURE

[Maximum marks: 100] Time : 3 hrs

PART – A

(Maximum Marks: 10)

I. Answer the following questions in one or two sentences. Each question carries 2 marks.

1. Write the meaning code of signal, both arms extended forward horizontally and depressed briskly.
2. Name the device used for orienting the plane table to magnetic north.
3. Write the least count of telescopic levelling staff.
4. Name the process of turning the telescope in vertical plane through 180^o about the trunnion axis.
5. Identify the instrument- a combination of electronic theodolite, measuring device and a microprocessor with memory unit.

. [5x2=10]

PART – B

(maximum mark :30)

II. Answer any five of the following. Each question carries 6 marks.

1. Differentiate between simple levelling and differential levelling.
2. Define ranging. Explain reciprocal ranging.
3. Determine the magnetic declination if the magnetic bearing of the sun at noon is $356^o42'$.
4. Explain about Remote sensing.
5. Convert the reduced bearing into whole circle bearing.
   I, S $32^o15'$ E   II, S $1^o0'$ W   III, N $12^o45'$ W
6. Give the classification of surveying.
7. Explain the method of reiteration for measuring horizontal angle. [5x6=30]

PART – C

(Answer one full question from each module. Each question carries 15 marks)

MODULE I

III a) Differentiate between back sight and fore sight. 6

b) Establish the method of chaining on uneven or sloping ground. 9

OR

IV a) Explain the different types of bench marks. 5

b) Plot the field and calculate the area. All measurements are in meters.

	752	D
	600	103 E
C 106	450	
	312	85 F
B 97	181	
	100	115 G
	0	
	A	

10

MODULE II

V. a) List the instruments used for levelling and explain their functions. 6

b) The following readings were taken using a levelling instrument. RL of bench mark =+100m staff reading at BM =.850, A= .900, Change point =1.005, .985, B= .940 and C= .930. Determine the reduced levels of A, B and C. 9

OR

VI The following consecutive readings are taken on a level with station A as bench mark. RL of the BM is 200.000m. 2.190, 3.150, 1.060, 0.230, 3.430, 3.170, 3.420, 3.720, 2.390. The instrument is shifted after the reading 3.430. Enter the readings in level book and calculate the reduced levels of all points. 15

MODULE III

V a) State face left and face right observations. 6

b) Explain the method of measuring the magnetic bearing of a line with a theodolite. 9

OR

VI a) List out the fundamental lines of a theodolite 5

b) The latitude and departure of a traverse ABCD are given as follows

Line	Latitude		Departure	
	N	S	E	W
AB	204.60		113.90	
BC		234.90	205.80	
CD		150.70		86.00
DA	181.00			233.70

Calculate the area if the sides are measured in meters by co-ordinate method 10

MODULE IV

IX a) List the application of GIS in Architecture. 5

b) Explain the procedure for measuring the area of a plot using total procedure. 10

OR

X a) Write the brief description about distomat. 5

b) Write the required steps for the initial setting of a total station for a field work. 10

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